Emerging Technology Trends: Insights from Stack Overflow Developer Survey 2024

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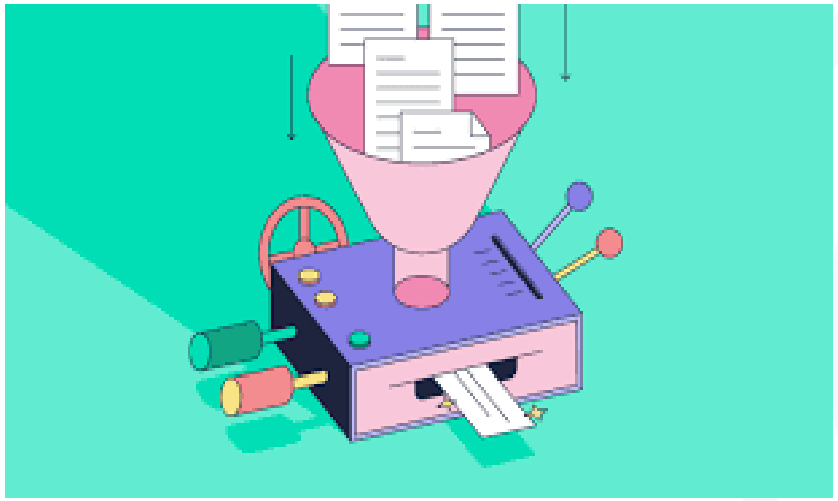
OUTLINE

- Executive Summary
- Introduction
- Methodology
- Results
  - Visualization – Charts
  - Dashboard
- Discussion
  - Findings & Implications
- Conclusion
- Appendix

EXECUTIVE SUMMARY



- HTML/CSS and JavaScript are the most widely used programming languages among developers
- PostgreSQL dominates database usage, followed by Microsoft SQL Server and MySQL
  - PostgreSQL is also the most desired database for future use
  - MongoDB remains popular for NoSQL needs
  - Cloud databases like Snowflake are gaining traction
- Amazon Web Services (AWS) and Google Cloud are the leading platforms used and desired by developers
- The majority of respondents (41.3%) are aged 25–34, with Bachelor's degrees being the most common education level
- React, Spring Boot and ASP.NET Core are the top desired web frameworks for the future

INTRODUCTION

Purpose of the report, Target Audience & Value of the analysis

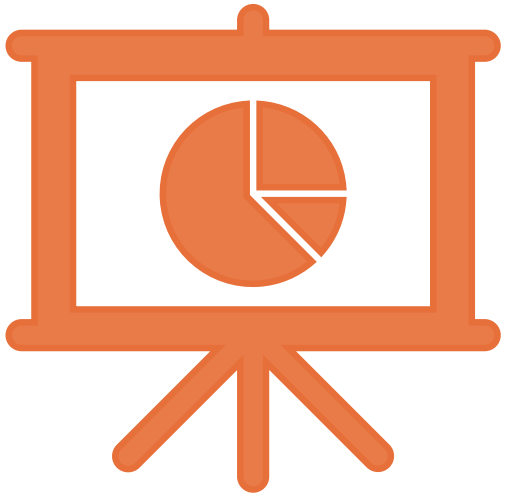
- This report analyzes the 2024 Stack Overflow Developer Survey to identify current and emerging technology trends
- Target audience: IT managers, business leaders, recruiters, and technology consultants seeking data-driven insights
- The analysis covers programming languages, databases, platforms, web frameworks, and developer demographics
- Value of the analysis:
  - Helps organizations identify in-demand skills for hiring and training decisions
  - Guides strategic technology investment by highlighting tools developers want to adopt

METHODOLOGY

Data sources, Data collection & Data wrangling steps

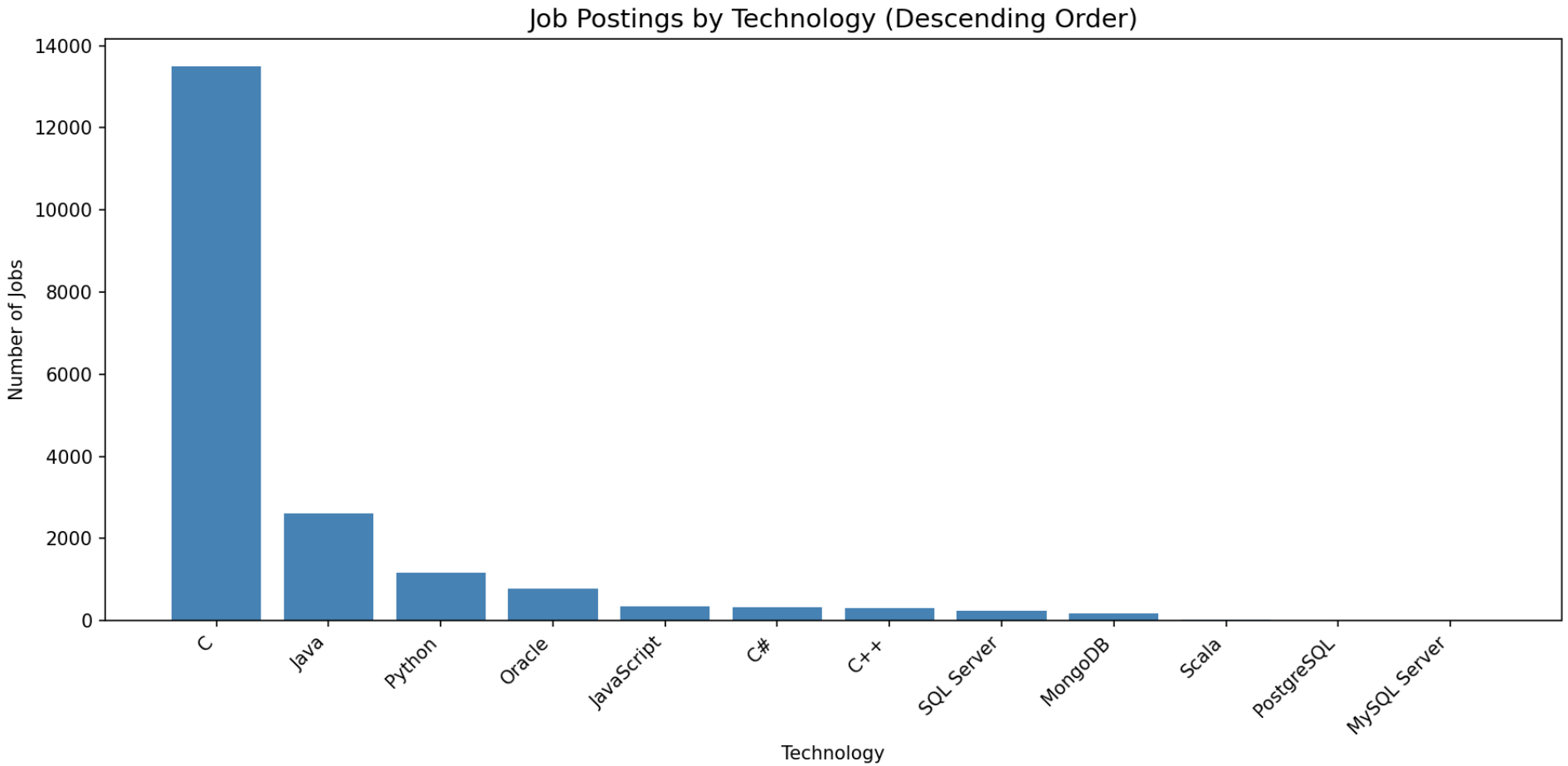
- Data Source: 2024 Stack Overflow Developer Survey provided as a pre-processed CSV file (`survey-data-updated.csv`) with 1,168 responses and 115 fields
- Data Collection: Gathered via API access and web scraping techniques using Python (BeautifulSoup, Requests)
- Data Wrangling: Cleaned dataset by removing duplicates, handling missing values, and normalizing data using Pandas and NumPy in Jupyter Notebook
- Analysis Tools Used:
  - Python (Pandas, NumPy, Matplotlib, Seaborn) for EDA and visualizations
  - Google Looker Studio for interactive dashboard creation

Results

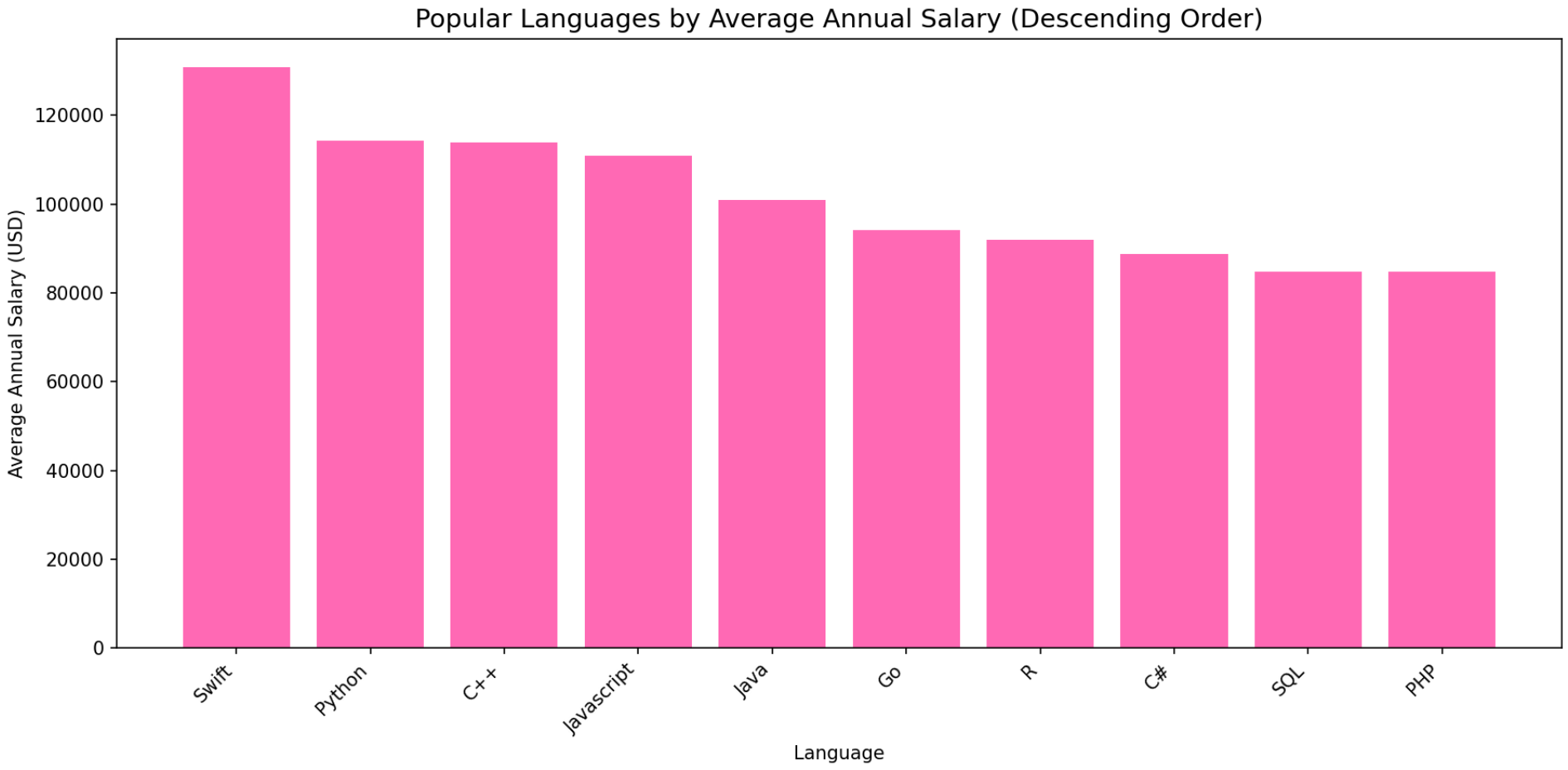


<Please present your results in the following slides.>

JOB POSTINGS



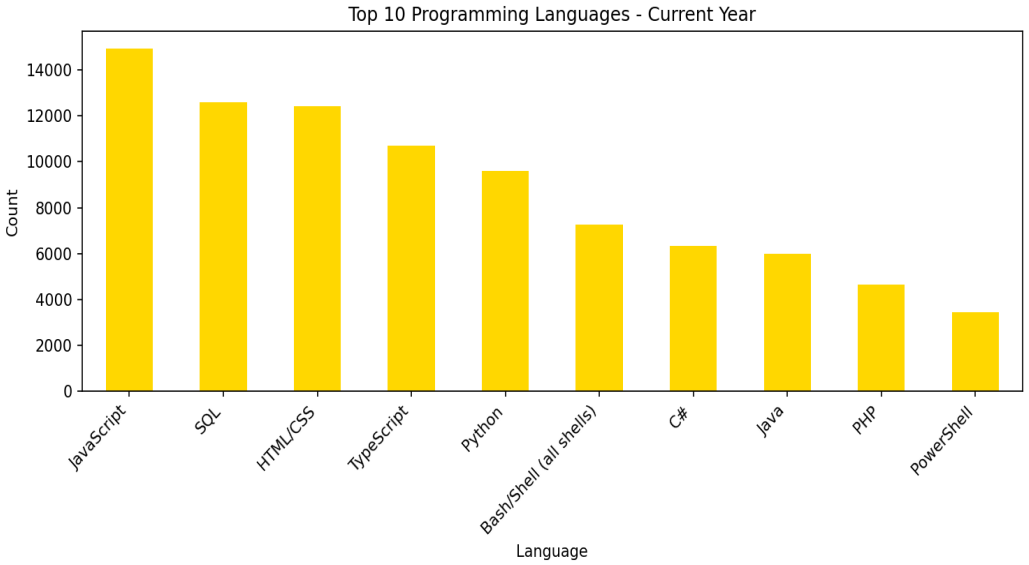
POPULAR LANGUAGES



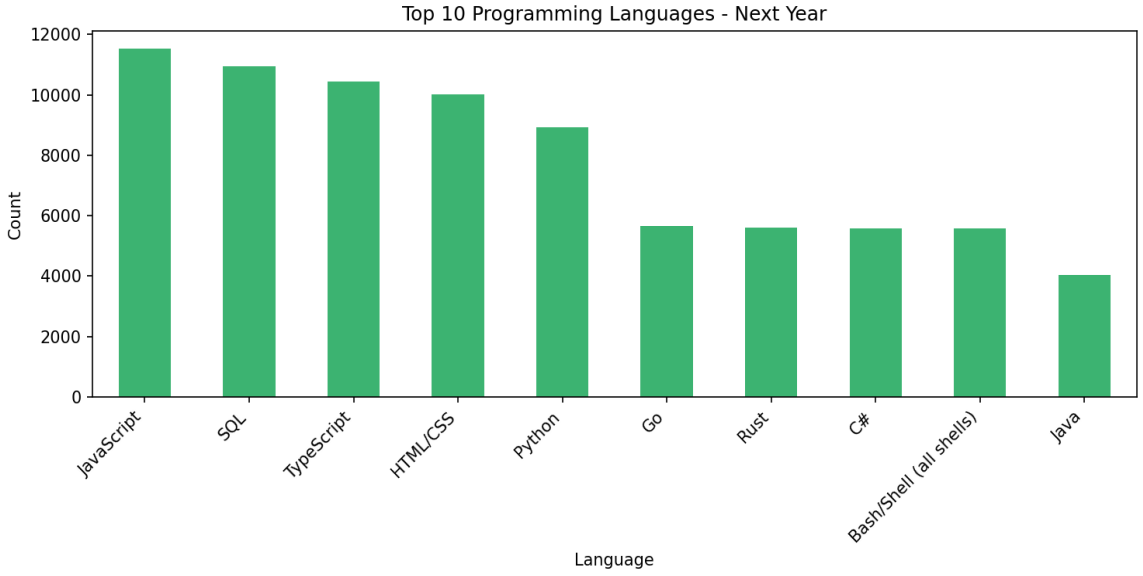
PROGRAMMING LANGUAGE TRENDS

Summarize key trends shown in the charts.

Current Year



Next Year



PROGRAMMING LANGUAGE TRENDS - FINDINGS & IMPLICATIONS

Findings

- JavaScript is the most used language currently (14,856 responses) and remains #1 desired for next year
- SQL and HTML/CSS are consistently in the top 3 for both current and future use
- Rust and Go are emerging as rising languages developers want to learn next year

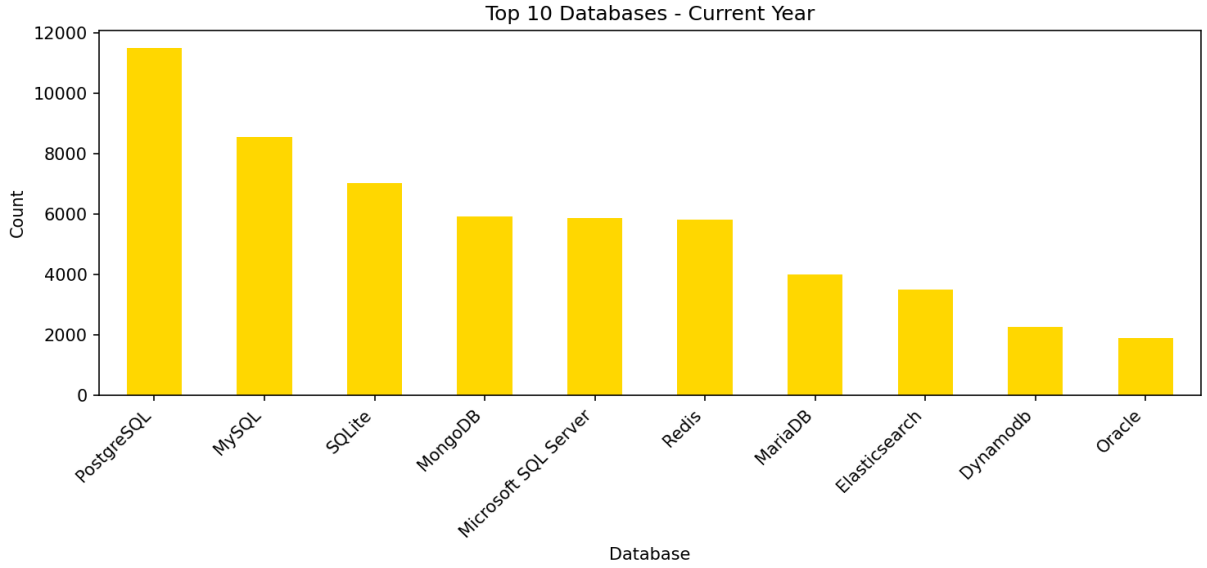
Implications

- Organizations should prioritize JavaScript, SQL, and HTML/CSS skills when hiring or training developers
- Full-stack development remains the dominant skill set demanded across the industry
- Investing in Rust and Go training now will prepare teams for future technology shifts

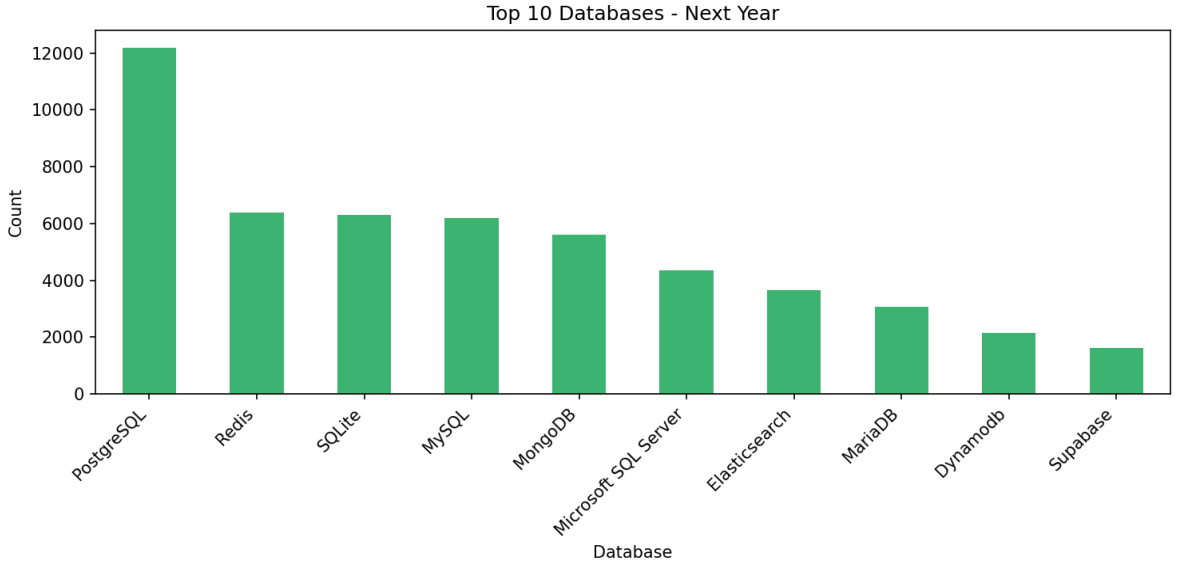
DATABASE TRENDS

Summarize key trends shown in the charts.

Current Year



Next Year



DATABASE TRENDS - FINDINGS & IMPLICATIONS

Findings

- PostgreSQL is the most used database currently (11,557 responses) and remains #1 desired for next year
- MySQL and SQLite are consistently in the top 3 for current usage, showing strong relational database preference
- Redis and Supabase are emerging as rising databases developers want to adopt next year

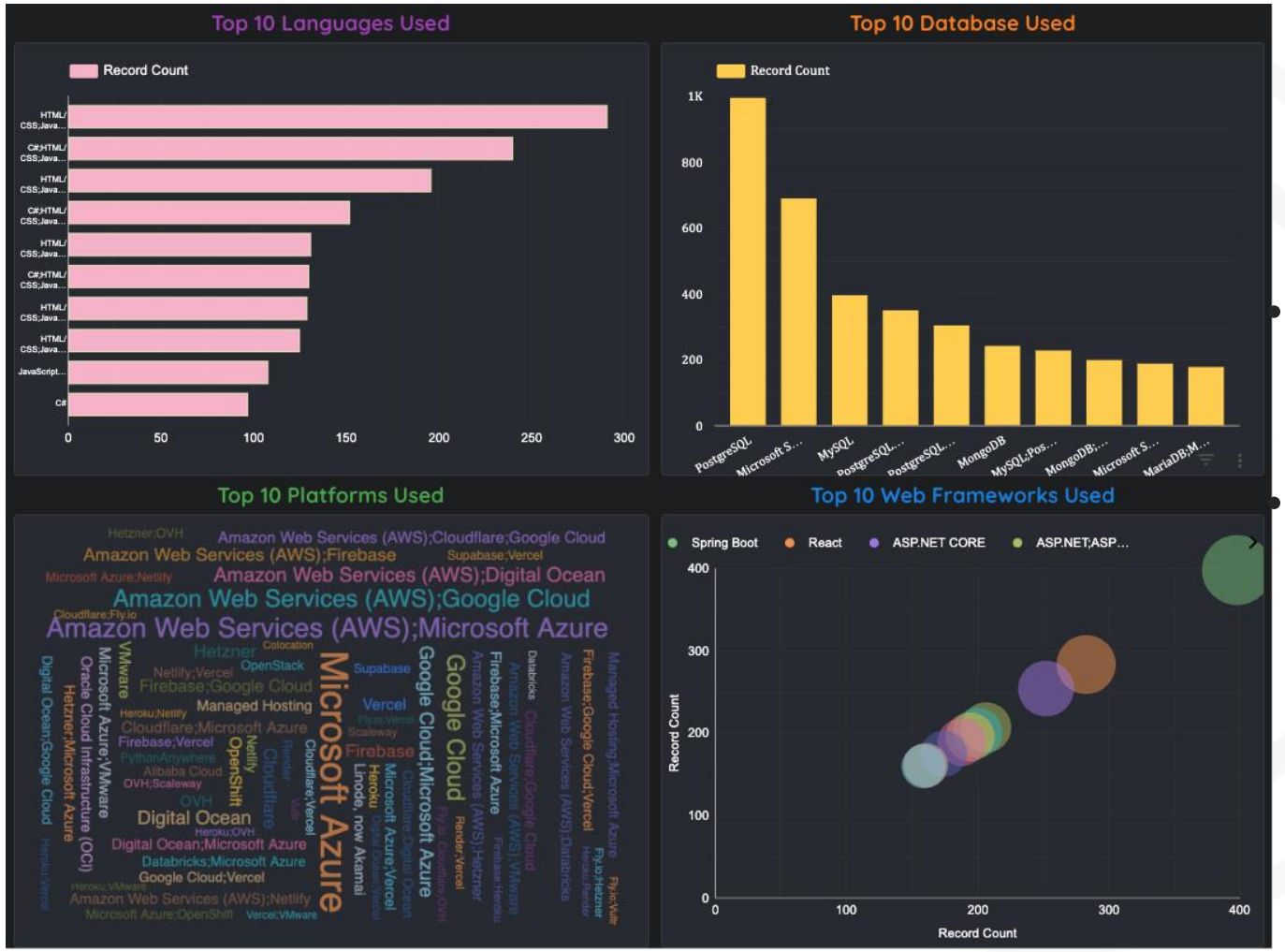
Implications

- PostgreSQL expertise is essential for developers — organizations should prioritize it in hiring and training
- Relational databases remain dominant, but NoSQL (MongoDB, Redis) is growing in demand
- Supabase's rise signals growing interest in open-source, developer-friendly database solutions

DASHBOARD

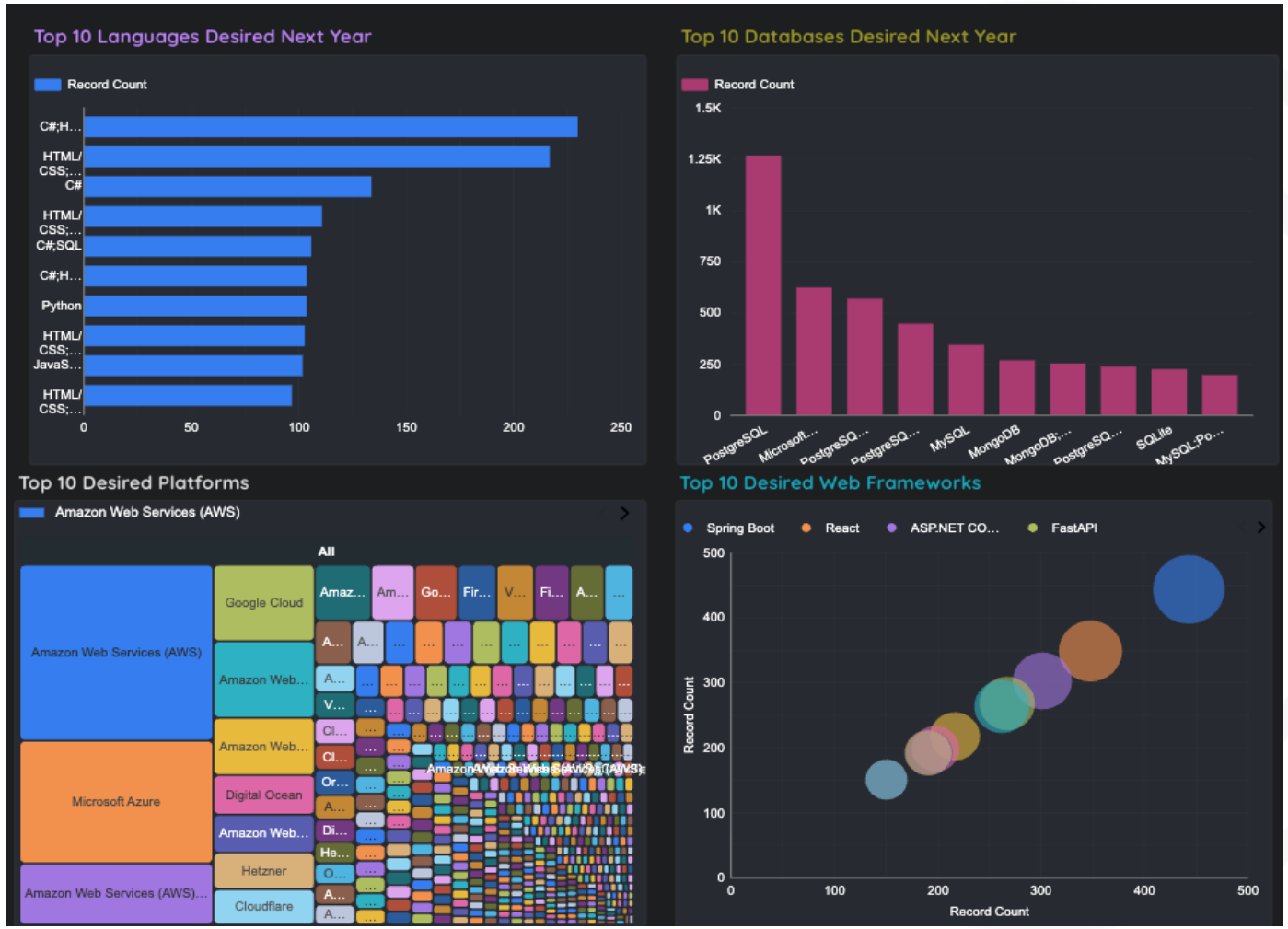
<Please present your dashboard in the following slides.>

DASHBOARD TAB 1:Current Technology Usage



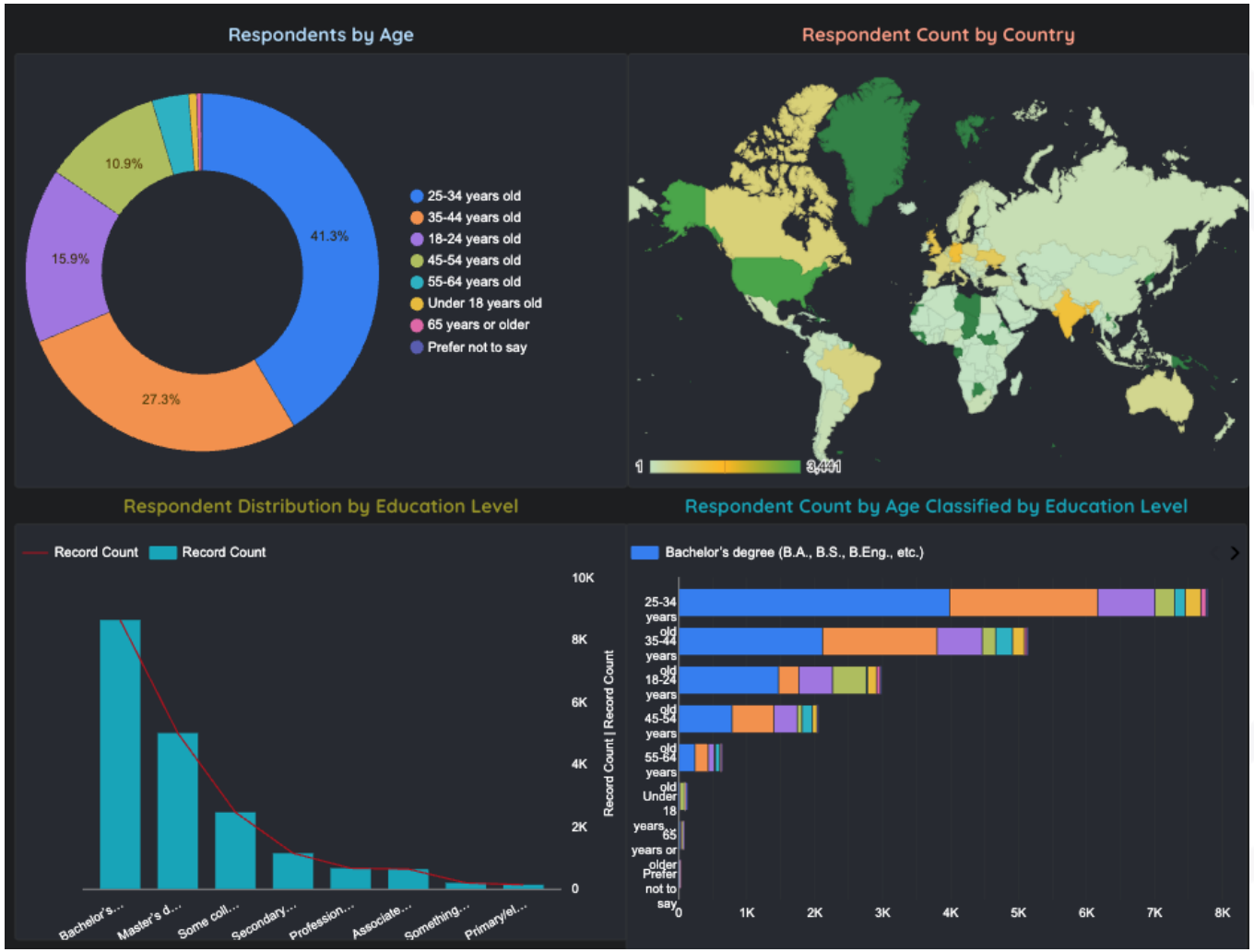
- Dashboard Insight: HTML/CSS and JavaScript dominate current language usage; PostgreSQL leads database usage
- Summarize key insights shown: AWS and Google Cloud are the top platforms; React and Spring Boot lead web frameworks

DASHBOARD TAB 2: Future Technology Trends



- **Dashboard Insight:** JavaScript and PostgreSQL remain the most desired technologies for next year, with Rust and Go gaining momentum
- **Summarize future trends:** AWS leads desired platforms; React and Spring Boot are the top desired web frameworks for next year

DASHBOARD TAB 3: Demographics



- **Dashboard Insight:** 41.3% of respondents are aged 25–34, with Bachelor's degree being the most common education level; USA and India lead in respondent count
- **Summarize demographic insights:** The developer community is predominantly young professionals (25–44), highly educated, and globally distributed across North America, Europe, and Asia

DISCUSSION

- JavaScript, SQL, and PostgreSQL consistently dominate both current usage and future aspirations, confirming their long-term relevance in the tech industry
- Emerging technologies like Rust, Go, and Supabase are gaining traction, signaling a shift toward performance-focused and developer-friendly tools
- The developer demographic is predominantly young (25–34), highly educated, and globally distributed — indicating a diverse, growing talent pool
- AWS and Google Cloud lead platform adoption, reflecting the continued dominance of cloud infrastructure in modern development
- Organizations should align hiring, training, and technology investment strategies with these trends to remain competitive

OVERALL FINDINGS & IMPLICATIONS

Findings

- JavaScript and PostgreSQL are the most used and most desired technologies across languages and databases
- AWS and Google Cloud dominate platform usage, with cloud adoption continuing to grow among developers
- The developer community is young (25–34), Bachelor's educated, and globally distributed across US, India, and Europe

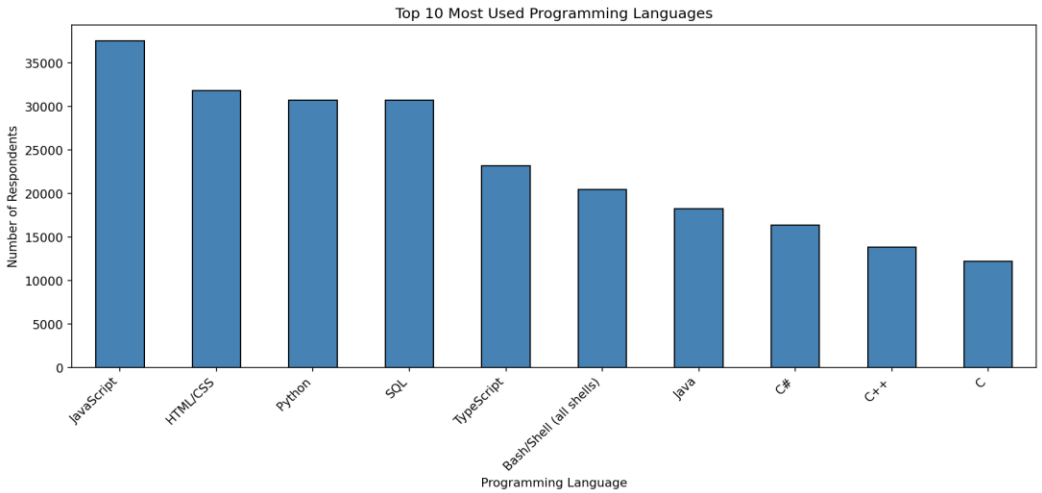
Implications

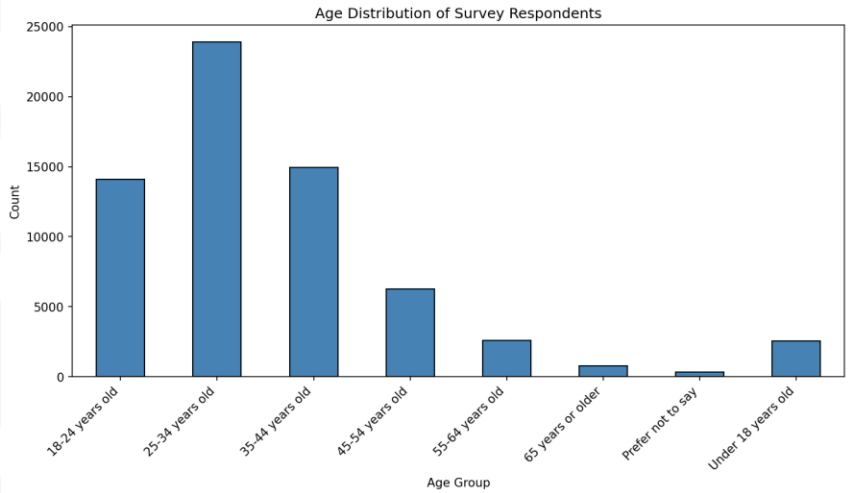
- Companies should prioritize JavaScript, Python, and PostgreSQL skills in recruitment and upskilling programs
- Cloud platform expertise (AWS, Google Cloud) is becoming a core requirement for modern developers
- Global diversity in the developer community presents opportunities for remote hiring and international collaboration

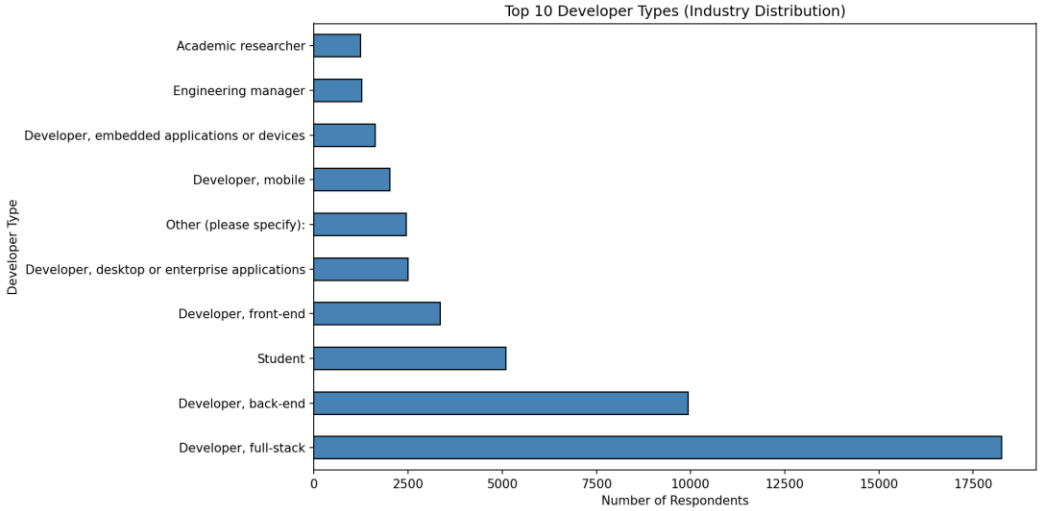
CONCLUSION

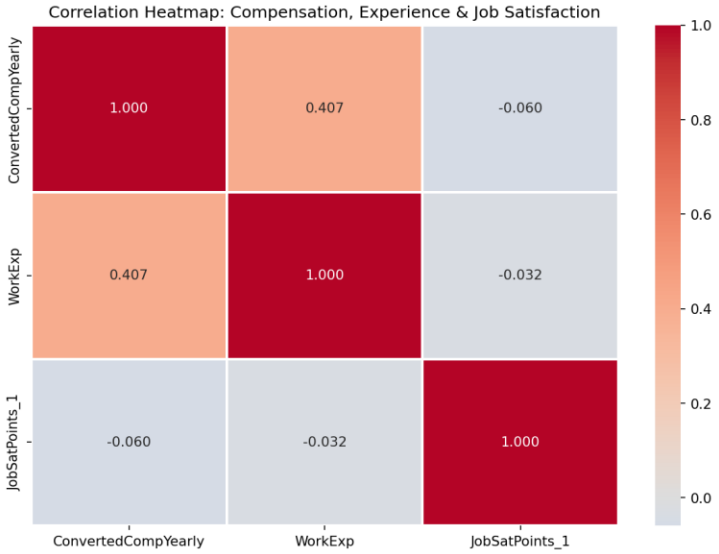
- This analysis of the 2024 Stack Overflow Developer Survey reveals that JavaScript, SQL, and PostgreSQL remain the backbone of modern development, with consistent demand across current and future usage
- Cloud platforms, particularly AWS and Google Cloud, are now essential infrastructure — developers must build cloud skills to stay competitive
- Emerging technologies like Rust, Go, and Supabase signal a shift toward performance, scalability, and developer-friendly tools that organizations should monitor and invest in
- The global, young, and highly educated developer demographic presents significant opportunities for organizations to build diverse, remote-friendly teams and invest in continuous learning programs

APPENDIX


Top 10 Most Used Programming Languages
Number of Respondents
0
5000
10000
15000
20000
25000
30000
35000
JavaScript
HTML/CSS
Python
SQL
TypeScript
Bash/Shell (all shells)
Java
C#
C++
C
Programming Language


Age Distribution of Survey Respondents
Count
0
5000
10000
15000
20000
25000
18-24 years old
25-34 years old
35-44 years old
45-54 years old
55-64 years old
65 years or older
Prefer not to say
Under 18 years old
Age Group


Top 10 Developer Types (Industry Distribution)
Developer Type
Academic researcher
Engineering manager
Developer, embedded applications or devices
Developer, mobile
Other (please specify):
Developer, desktop or enterprise applications
Developer, front-end
Student
Developer, back-end
Developer, full-stack
0
2500
5000
7500
10000
12500
15000
17500
Number of Respondents


Correlation Heatmap: Compensation, Experience & Job Satisfaction
ConvertedCompYearly
WorkExp
JobSatPoints_1
1.000
0.407
-0.060
0.407
1.000
-0.032
-0.060
-0.032
1.000
ConvertedCompYearly
WorkExp
JobSatPoints_1
1.0
0.8
0.6
0.4
0.2
0.0